

Engineering Mathematics Study Notes: Advanced Application of Modified Euler's Theorem

This note breaks down the step-by-step solution for a complex second-order differential operator involving a radical and an inverse cosecant wrapper function. We will utilize **Modified Euler's Theorem** to resolve the expression cleanly without resorting to direct partial differentiation.

Problem Statement

If $u = \csc^{-1} \sqrt{\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}}$, prove that:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144} (13 + \tan^2 u)$$

Complete Step-by-Step Derivation

Step 1: Transform the Function to Remove the Non-Linear Wrapper

The function u is not homogeneous in its raw form due to the outer inverse cosecant function and the square root. We eliminate these step-by-step by taking the cosecant of both sides and squaring the result to isolate a core algebraic function z :

$$\begin{aligned} \csc u &= \sqrt{\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}} \\ \csc^2 u &= \frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \end{aligned}$$

Let our new homogeneous substitute function be z :

$$z = \csc^2 u = \frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}$$

Step 2: Determine the Degree of Homogeneity (n)

Let the pure algebraic fraction be represented as $f(x, y)$:

$$f(x, y) = \frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}}$$

To determine the degree of homogeneity, substitute $x \rightarrow xt$ and $y \rightarrow yt$:

$$f(xt, yt) = \frac{(xt)^{1/2} + (yt)^{1/2}}{(xt)^{1/3} + (yt)^{1/3}}$$

Factor out the scaling parameter t from both the numerator and the denominator:

$$f(xt, yt) = \frac{t^{1/2}(x^{1/2} + y^{1/2})}{t^{1/3}(x^{1/3} + y^{1/3})}$$

Simplify the exponents of t using the laws of indices:

$$f(xt, yt) = t^{(\frac{1}{2} - \frac{1}{3})} \left[\frac{x^{1/2} + y^{1/2}}{x^{1/3} + y^{1/3}} \right]$$

Calculate the common denominator for the exponent:

$$\frac{1}{2} - \frac{1}{3} = \frac{3 - 2}{6} = \frac{1}{6}$$

$$f(xt, yt) = t^{1/6} f(x, y)$$

Since t factors out with a power of $1/6$, the substitute function z is a homogeneous function with a degree of $n = \frac{1}{6}$.

Step 3: Compute the First-Order Value Expression $g(u)$

According to the Modified Euler's Theorem, if a function is transformed via $z = f(u)$, the first-order partial differential expression simplifies to a single-variable function $g(u)$:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n \cdot \frac{f(u)}{f'(u)} = g(u)$$

From our setup:

$$f(u) = \csc^2 u$$

Differentiate $f(u)$ with respect to u using the chain rule:

$$f'(u) = 2 \csc u \cdot (-\csc u \cot u) = -2 \csc^2 u \cot u$$

Substitute $n = \frac{1}{6}$, $f(u)$, and $f'(u)$ into the formula to evaluate $g(u)$:

$$g(u) = \frac{1}{6} \cdot \frac{\csc^2 u}{-2 \csc^2 u \cot u}$$

Cancel out the common $\csc^2 u$ term from the numerator and denominator:

$$g(u) = \frac{1}{6} \cdot \frac{1}{-2 \cot u}$$

$$g(u) = -\frac{1}{12 \cot u}$$

Using the trigonometric identity $\frac{1}{\cot u} = \tan u$:

$$g(u) = -\frac{1}{12} \tan u$$

Step 4: Apply the Second-Order Modified Euler's Corollary

The second-order differential operator for a modified homogeneous function is determined by the following expansion identity:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = g(u) \cdot [g'(u) - 1]$$

Find the derivative of $g(u)$ with respect to u :

$$g'(u) = \frac{d}{du} \left(-\frac{1}{12} \tan u \right) = -\frac{1}{12} \sec^2 u$$

Substitute the expressions for $g(u)$ and $g'(u)$ into the right-hand side (RHS) of the second-order corollary:

$$\text{RHS} = \left(-\frac{1}{12} \tan u \right) \cdot \left[-\frac{1}{12} \sec^2 u - 1 \right]$$

Step 5: Expand and Simplify to Match the Target Identity

Factor out the negative sign from the expression inside the brackets to cancel the leading negative sign:

$$\begin{aligned} \text{RHS} &= \left(-\frac{1}{12} \tan u \right) \cdot (-1) \cdot \left[\frac{1}{12} \sec^2 u + 1 \right] \\ \text{RHS} &= \frac{1}{12} \tan u \cdot \left[\frac{\sec^2 u + 12}{12} \right] \end{aligned}$$

Multiply the denominators together outside the bracket:

$$\text{RHS} = \frac{\tan u}{144} \cdot [\sec^2 u + 12]$$

To match the target equation format, convert $\sec^2 u$ to its tangent equivalent using the standard Pythagorean identity $\sec^2 u = 1 + \tan^2 u$:

$$\text{RHS} = \frac{\tan u}{144} \cdot [(1 + \tan^2 u) + 12]$$

Combine the constant numerical values inside the bracket:

$$\text{RHS} = \frac{\tan u}{144} \cdot [13 + \tan^2 u]$$

Equating this fully simplified right-hand side back to the second-order differential operator completes the verification:

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = \frac{\tan u}{144} (13 + \tan^2 u)$$

Hence, proved.